

Vaibhav Singh

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Relevant Research & Engineering Experience

DV CAD Intern, Xorvis Inc.

March '26 – May '26

AI-Driven Formal Verification & EDA Infrastructure | Python, LangGraph, SystemVerilog

Newark, DE, US (Remote)

- **Automated Formal Verification:** Engineered an **agentic workflow** to autonomously detect functional mismatches between natural language specifications and RTL code.
- **Semantic Extraction Pipeline:** Normalized specs & RTL into unified **Control Data Flow Graph** for semantic comparison.
- **Assertion Synthesis:** Generated targeted **SystemVerilog Assertions (SVA)** directly from identified CDFG discrepancies.

STA CAD Engineer (Contract), Silimate (YC S23)

Nov '25 – Feb '26

EDA Toolchain Optimization & Validation | C++, Tcl, Python, CMake

Mountain View, CA, US (Remote)

- **Liberty File Optimization:** Built a C++ **Visitor-based CCS & SI pruning engine** for Liberty (.lib) files, achieving a **75% size reduction** to accelerate EDA tool loads.
- **Compiler Validation Pipeline:** Integrated **VlogHammer**, **Chimera**, and **Verismith** into **sv-tests**, uncovering critical edge-case failures in Verilog & SystemVerilog for EDA tools.

Undergraduate Researcher, Nanoscale Devices and Circuits Lab

Jan '25 – Dec '25

MIS Modeling Extension in OpenSTA | ML, C++, OpenSTA, SPICE, Python

IIITD, Delhi, India

- **ML-Enhanced Timing Analysis:** Upgraded OpenSTA to accurately model realistic **7nm Multi-Input Switching (MIS)** behavior, recovering critical timing margins lost by standard Single-Input Switching (SIS) approximations.
- **C++ Core Integration:** Modified **OpenSTA's source code** to bypass standard liberty NLDM look-up tables, directly embedding trained **ANN/RF inference engines** into the delay calculation path.
- **Simulation Performance:** Trained models on SPICE simulation data to achieve a **10,000x speedup** with just 5% error, mathematically recovering **27.8% delay pessimism** and **3.9% slack**.

Undergraduate Engineer, Advance Multicore System Lab

Jan '25 – Aug '25

RISC-V Micro-controller for Edge AI Computing | RISC-V, SystemVerilog, FPGA, Synthesis, STA

IIITD, Delhi, India

- **Coprocessor Integration:** Integrated a custom **vector arithmetic coprocessor** with the **CV32E40X RISC-V core** using the **XIF interface**, validating functionality via RTL simulation and **FPGA** deployment.
- **RTL-to-GDSII Flow:** Executed the physical implementation for the SoC, managing Logic Synthesis and Physical Design.

Education

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)

2022 – 2026

B.Tech in Electronics and VLSI Engineering CGPA: 7 (Major GPA: 8.5)

Delhi, India

Relevant Projects

EDA_MCP: AI Agent Bridge for Remote EDA Automation [\[Link\]](#) | Python, MCP, SSH, Agentic Tooling

July '26 – Present

- **Autonomous EDA Orchestration:** Engineered an **AI co-development platform** that fully automates design tasks like **writing scripts**, executing tools (**Genus**, **Tempus**, **Virtuoso**), and **fixing bugs** to fulfill engineer requests—leveraging a **knowledge-based mind map** to retrieve design-specific context.
- **State Synchronization Engine:** Built a **Git-backed** workspace pipeline featuring **auto-commit baselines**, **unified diffing**, and sub-10ms **SSH multiplexed transport**, providing agents real-time access to scripts, logs, waveforms and other files.

CPU-Memory Architectural Modeling [\[Link\]](#) | SystemC, TLM 2.0, C++

Aug '25 – Sep '25

- **Dual-Abstraction Prototyping:** Designed a cycle-approximate **SystemC TLM 2.0** model featuring **loosely-timed** blocking loops for SW-to-CPU and **approximate-timed** non-blocking interfaces for CPU-to-Memory.
- **Microarchitectural Simulation:** Modeled discrete latencies, back-pressure mechanisms, and transaction-phase callbacks (**BEGIN_RESP**) to simulate memory structural hazards and concurrent HW/SW execution loops.

Technical Skills

Languages & HDL: SystemVerilog, Verilog, C++, Python, Tcl, Bash, SystemC

Domain Knowledge: Static Timing Analysis (STA), Timing Closure, Physical Design, SVA, UVM, Formal Verification, Library Modeling (.lib, NLDM, CCS, SI), Logic Synthesis, High-Level Synthesis (HLS)

Tools & Technologies: Cadence (JasperGold, Virtuoso, Genus, Tempus), OpenSTA, Yosys, Vivado, Git, Linux, CMake

AI & Machine Learning: Agentic AI, RAG, LLM, Reinforcement Learning (RL), Supervised Learning (ANN, RF, SVM)

Relevant Coursework: Advanced Digital Design and Verification, VLSI Design Flow, Digital Circuits, Analog CMOS, Embedded Logic Design, Integrated Electronics, Machine Learning, Data Structures & Algorithms (C++)

Problem Solving: Codeforces [Specialist](#), LeetCode (DSA) + HDLBits (Verilog) 300+ Solved